

Unified Privacy Guarantees for Decentralized Learning via Matrix Factorization

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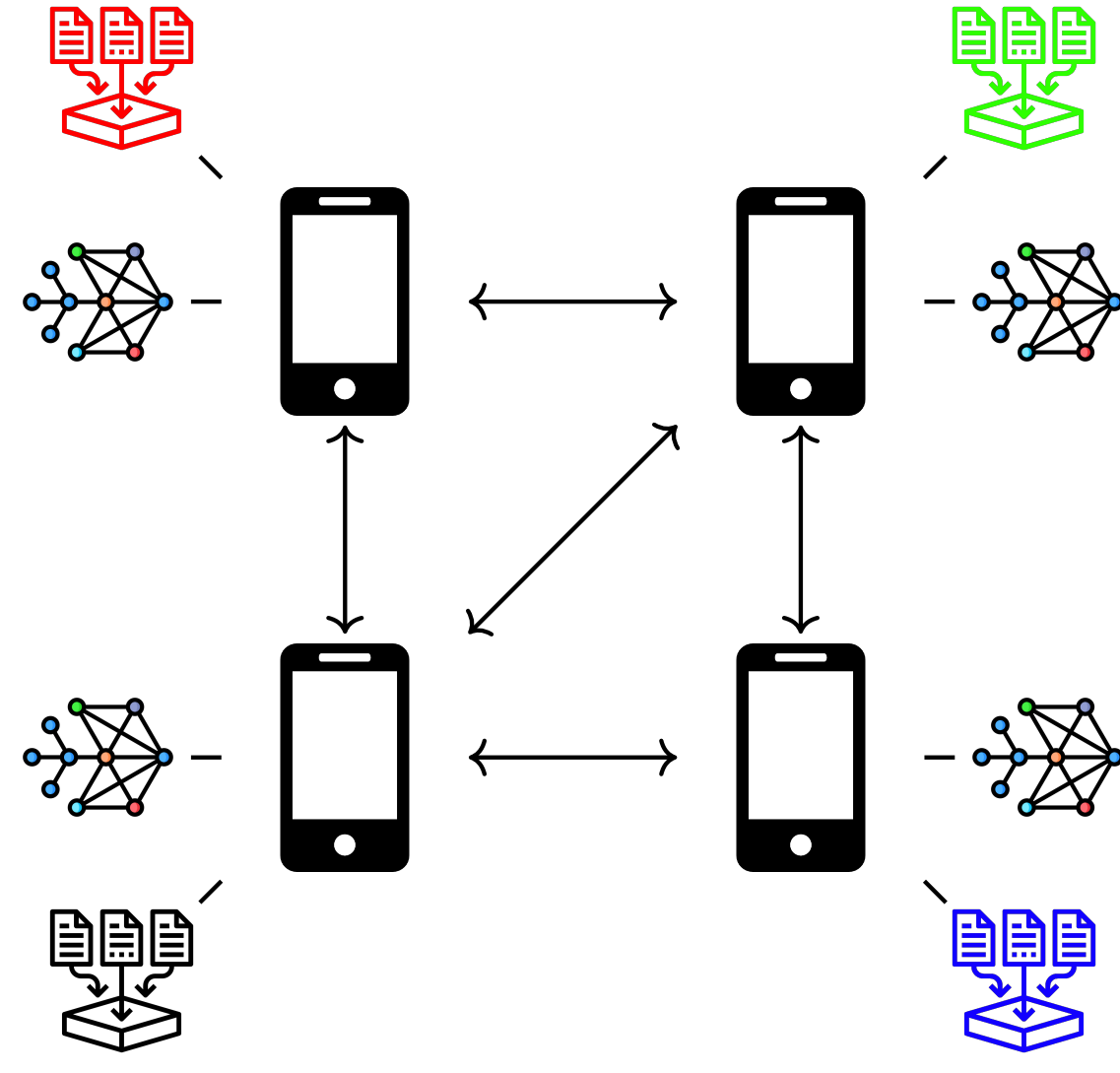
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Background

- Fixed undirected connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, n nodes; doubly stochastic gossip matrix W ($W_{uv} > 0$ iff $\{u, v\} \in \mathcal{E}$)
- Local objective: $f(\theta) = \sum_{v=1}^n f_v(\theta, D_v)$, D_v local dataset; local gradient $g_t^v = \nabla f_v(\theta_t^v, \xi_t^v)$
- D-SGD**: gradient step $\theta_{t+\frac{1}{2}} = \theta_t - \eta G_t$, gossip step $\theta_{t+1} = W\theta_{t+\frac{1}{2}}$, where G_t is the stacked gradient
- D-SGD is **vulnerable**: a node can reconstruct data even from nodes **far away** in the graph [2]
- Applying DP noise amplifies privacy through noise accumulation [1], but **no existing analysis handles correlated noise between nodes**



MF-D-SGD: Correlated Noise in D-SGD

Consider a general class of algorithms with **arbitrary** noise correlation:

Algorithm 1: MF-D-SGD: Matrix Factorization Decentralized SGD

Inputs: $W \in \mathbb{R}^{n \times n}$, $C, T, \Delta_g, \sigma, \theta_0 \in \mathbb{R}^{n \times d}$, $Z \sim \mathcal{N}(0, \Delta_g^2 \sigma^2)^{nT \times d}$

forall node u **in parallel** **do**

for $t = 1$ **to** T **do**

$g_t^{(u)} \leftarrow \text{clip}[\nabla f_u(\theta_t^{(u)}, \xi_t^{(u)}), \Delta_g]$ // Clipped gradient

$\theta_{t+\frac{1}{2}}^{(u)} \leftarrow \theta_t^{(u)} - \eta(g_t^{(u)} + (C^\dagger Z)_{[nt+u]})$ // Update with **correlated** noise

 Send $\theta_{t+\frac{1}{2}}^{(u)}$ to neighbors $v \in \Gamma_u$; receive $\theta_{t+\frac{1}{2}}^{(v)}$;

$\theta_{t+1}^{(u)} \leftarrow \sum_{v \in \Gamma_u} W_{[u,v]} \theta_{t+\frac{1}{2}}^{(v)}$ // **Local average**

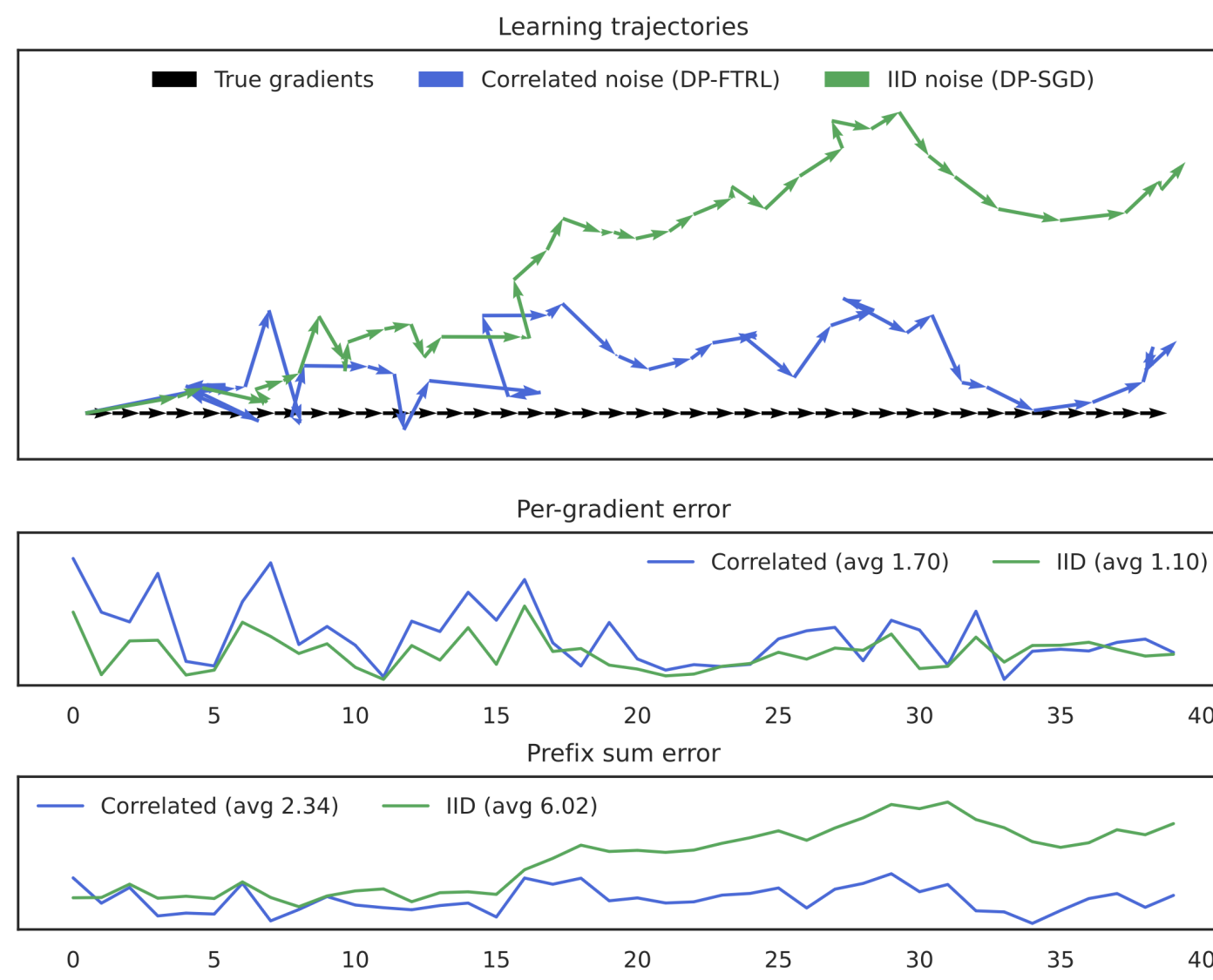
return $\theta_{T+1}^{(u)}$, $\forall u \in \{1, \dots, n\}$

Choice of C is (almost) free, which allows **arbitrary** spatial and temporal noise correlation, recovering all existing DP-D-SGD variants as special cases.

DP via Matrix Factorization

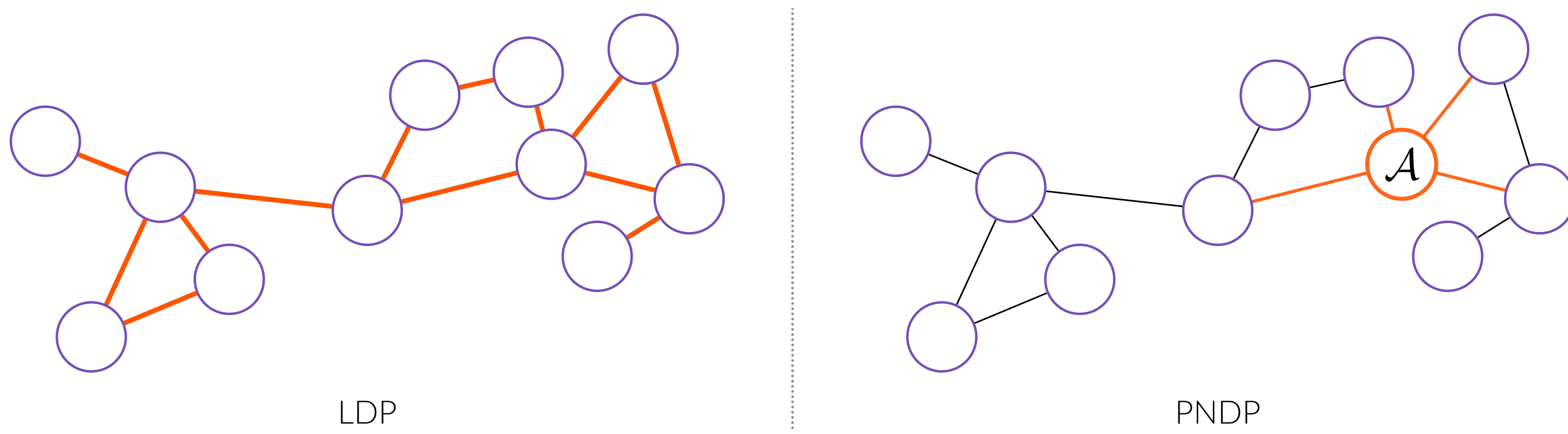
$\mathcal{M}$ satisfies μ -GDP if for any $\mathcal{D} \simeq \mathcal{D}'$ there exists h such that $h(Z) \stackrel{d}{=} \mathcal{M}(\mathcal{D})$, $Z \sim \mathcal{N}(0, 1)$ and $h(Z') \stackrel{d}{=} \mathcal{M}(\mathcal{D}')$, $Z' \sim \mathcal{N}(\mu, 1)$.

- Noise **accumulates** across iterations; correlated noise reduces total amount injected
- For $A_{i,j}^{\text{pre}} = \mathbf{1}_{i \geq j}$, DP-SGD writes as:
 $\theta = \mathbf{1}_T \otimes \theta_0 - (A^{\text{pre}} G + Z)$, $G \in \mathbb{R}^{T \times d}$
- For any factorization $A^{\text{pre}} = BC$, rewrite $A^{\text{pre}} G + BZ$ as $A^{\text{pre}}(G + C^\dagger Z)$: the mechanism now adds **structured** noise $C^\dagger Z$ to each gradient before the workload A^{pre} is applied
- Goal: minimize $\text{sens}(C)^2 \|B\|^2$ with
 $\text{sens}_\Pi(C) = \max_{G \succeq \Pi G} \|C(G - G')\|_F$
- Smaller sensitivity $\Rightarrow$ less noise needed for the same μ -GDP guarantee



Correlated (right) vs. i.i.d. noise (left).

Attacker Observations

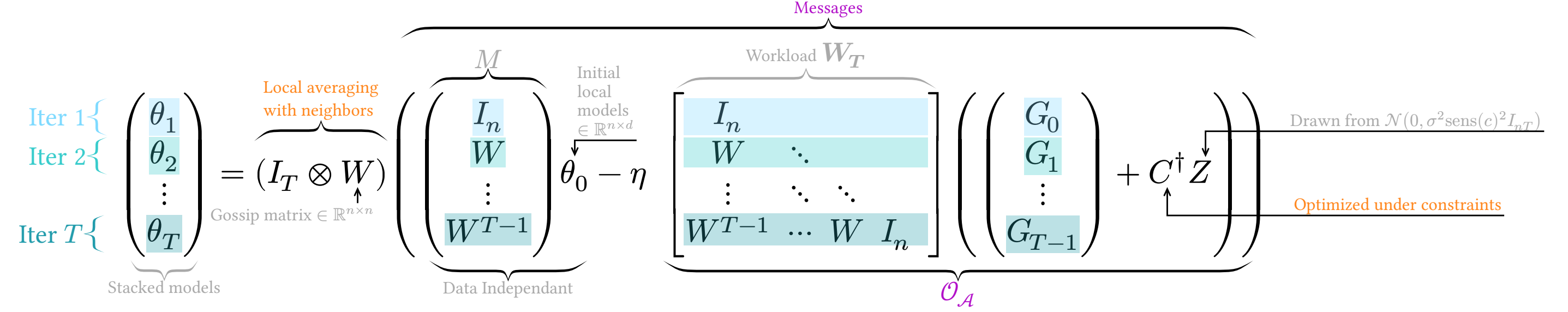


- $(M\theta_0 - \eta \mathbf{W}_T(G + C^\dagger Z))$ is the set of all messages **shared** during training
- Attacker observations $\mathcal{O}_A = AG + BZ$ are a post-processing of this set
- Captures different trust models: LDP, PNDP [1], SecLDP

References

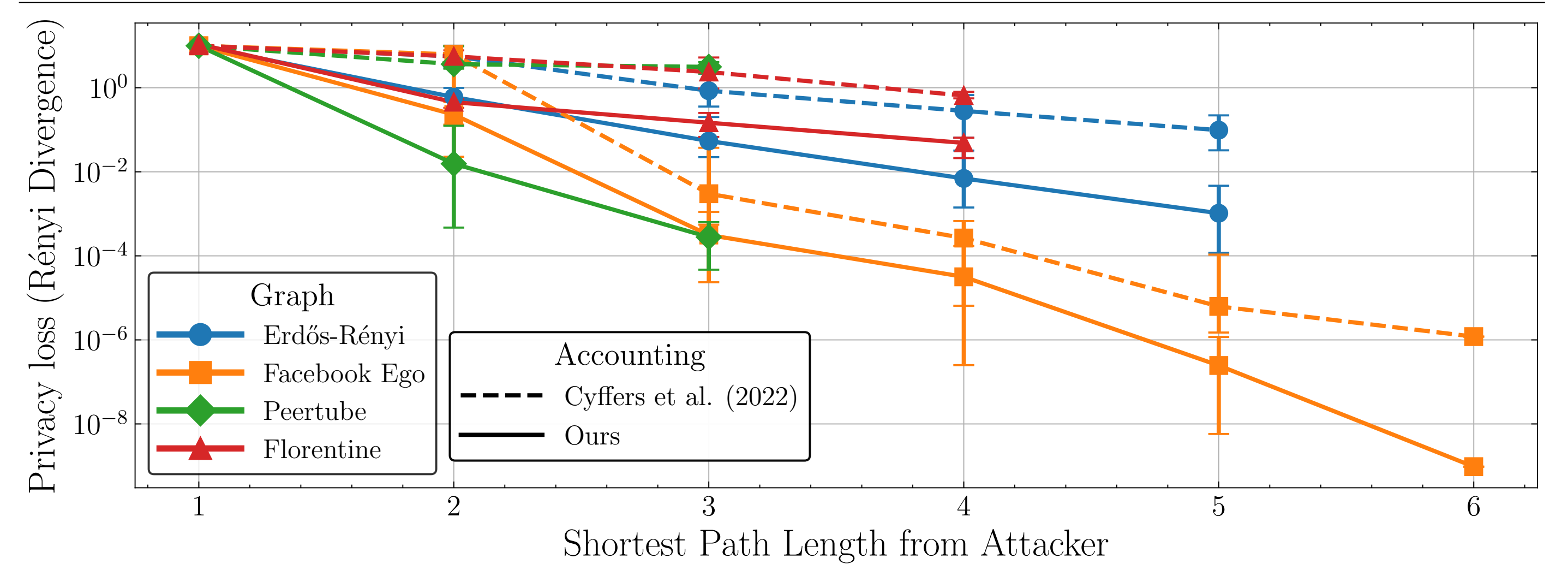
- [1] E. Cyffers, M. Even, A. Bellet, and L. Massoulié, *Muffliato: Peer-to-Peer Privacy Amplification for Decentralized Optimization and Averaging*, NeurIPS, 2022.
- [2] A. El Mrini, E. Cyffers, and A. Bellet, *Privacy attacks in decentralized learning*, ICML'24, JMLR.org, 2024.
- [3] K. Pillutla, J. Upadhyay, C. A. Choquette-Choo, K. Dvijotham, A. Ganesh, M. Henzinger, J. Katz, R. McKenna, H. B. McMahan, K. Rush, T. Steinke, and A. Thakurta, *Correlated noise mechanisms for differentially private learning*, 2025.

Framing MF-D-SGD as a Matrix Factorization Problem



- Full algorithm: $\theta = (I_T \otimes W)(M\theta_0 - \eta \mathbf{W}_T(G + C^\dagger Z))$
- Attacker observations $\mathcal{O}_A = AG + BZ$ are a post-processing of all shared messages; captures LDP, PNDP [1], SecLDP trust models

Better Accounting



Unified Framework for Privacy Guarantees

Theorem

Let $\mathcal{O}_A = AG + BZ$ be the attacker's knowledge and Π a participation scheme. For $Z \sim \mathcal{N}(0, \nu^2)^{m \times d}$, when A is column-echelon and $A = BC$ with

$$\nu = \sigma \text{sens}_\Pi(C; B), \quad \text{sens}_\Pi(C; B) \leq \max_{\pi \in \Pi} \sum_{s,t \in \pi} \left| (C^\top B^\dagger BC)_{s,t} \right|,$$

then $\mathcal{M}$ is $\frac{1}{\sigma}$ -GDP, even when G is chosen adaptively.

- Generalises prior results; dependency on B arises because the workload matrix is not square
- Holds for more general algorithms than MF-D-SGD

MAFALDA

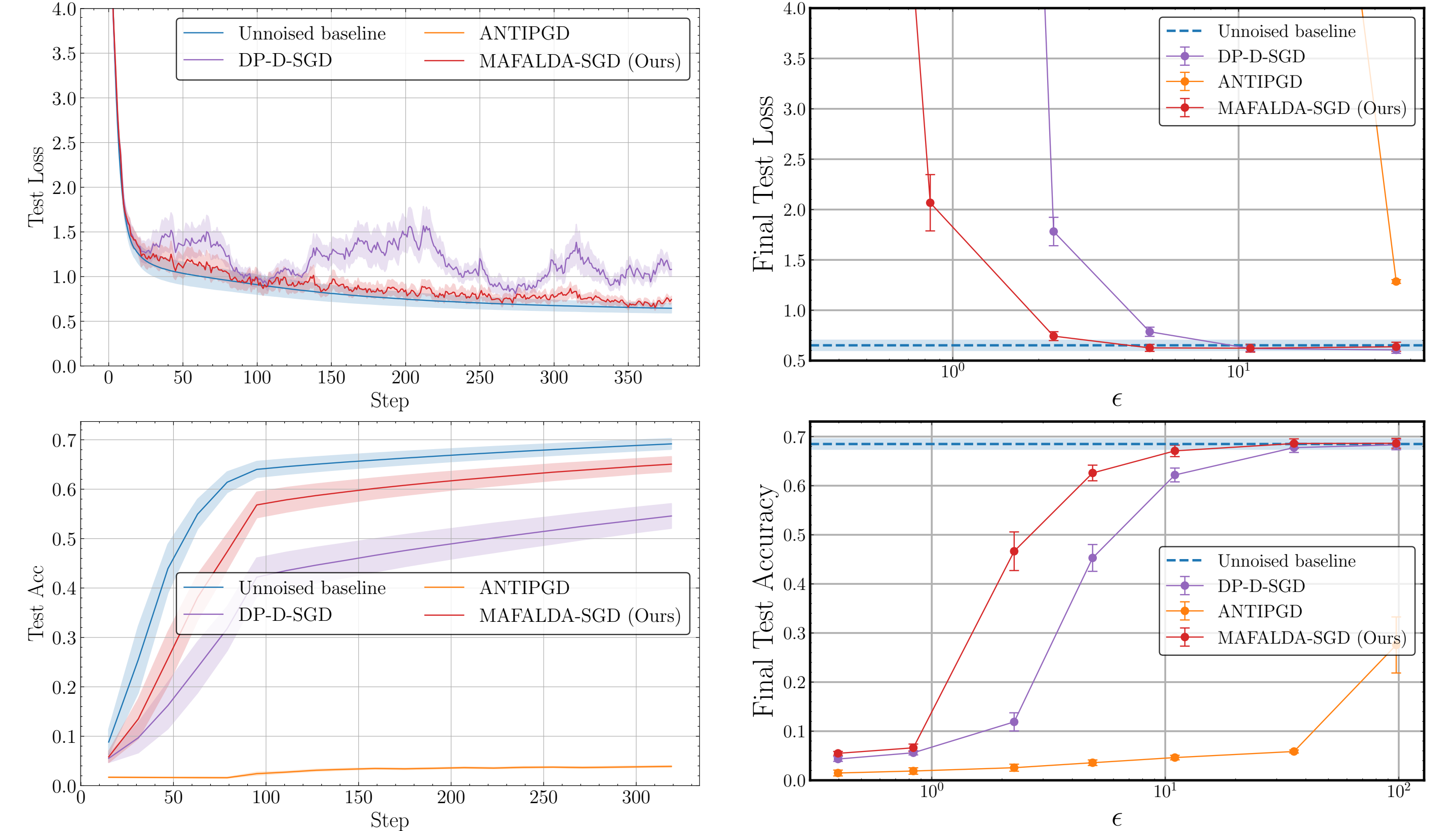
MAtrix **F**actorization for **L**ocal **D**ifferential PrivAte-SGD (MAFALDA-SGD) optimises C for Local DP (each node relies only on its own noise). Minimization objective:

$$\mathcal{L}_{\text{opti}}(\mathbf{W}_T, C_{\text{local}}) = \text{sens}_{\Pi_{\text{local}}}(C_{\text{local}})^2 \|LC_{\text{local}}^\dagger\|_F^2$$

where L is the Cholesky factor of

$$\sum_{i=1}^n \left[(I_T \otimes W) \mathbf{W}_T \mathbf{K}^{(T,n)} \right]_{[:,iT:(i+1)T-1]}^\top \left[(I_T \otimes W) \mathbf{W}_T \mathbf{K}^{(T,n)} \right]_{[:,iT:(i+1)T-1]}$$

Experiments on Housing and FEMNIST datasets (Facebook Ego graph):



Links



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